

TCP INVESTIGATION

During a capture review of my personal project, a TCP conversation on port 80 [140.248.134.172 – 192.168.0.53] was observed transferring large amounts of data with no HTTP visible in the packet view. This raised questions of whether this could be tunnelled traffic or something to investigate.

I followed the TCP stream *tcp.stream eq 94* which revealed a legitimate *HTTP/1.1 206 Partial content* response from *tlu.dl.delivery.mp.microsoft.com*. The Content Disposition header identified the payload as *Microsoft.YourPhone_1.26072.255.0_neutral_~_8wekyb3d8bbwe.AppxBundle* as an update package for the Phone link app delivered via Microsoft's Delivery Optimization. Headers such as *MS-CV*, *MS-CorrelationID* and *Cache-Control: public, max-age=17280000* are consistent with the Routine Microsoft Store/Windows Update traffic. The "encrypted-looking" appearance showed it was not evidence of tunnelling.

This broadens the view beyond a single stream which confirms pattern was a capture-wide, with dozens of *.cab* delta packages from *download.windowsupdate.com*, additional *octet-stream* objects from *dl.delivery.mp.microsoft.com/tlu.dl.delivery.mp.microsoft.com*, and *application/x-chrome-extension* objects from *msedge.b.tlu.delivery.mp.microsoft.com*. This confirms the capture had a routine OS and browser background and maintenance traffic.

The I/O graph showed bursty traffic with distinct spikes of up to 1 900 packets per second aligning with the large download activity which was interspersed with quieter periods.

Out of 95 118 total packets, 3.8% (3 581) were flagged by TCP analysis – predominantly duplicate ACKs, out of order segments and "previous segment not captured" notices, concentrated during the busiest traffic windows. No zero-window conditions or sustained retransmission storm were observed, and several flagged packets weren't errors at all

Wireshark - Follow TCP Stream (tcp.stream eq 94) - Wi-Fi

00000180 6f 6e 74 65 6e 74 2d 4c 65 6e 67 74 68 3a 20 30 content-Length: 0
00000180 6d 6a 48 6f 73 74 3a 20 74 6c 75 2e 64 6c 2e 64 ..Host: tlu.dl.d
00000180 65 6c 69 76 65 72 79 2e 6d 70 2e 6d 69 63 72 6f elivery.mp.micro
00000180 73 6f 66 74 2e 63 6f 6d 0d 0a 0d 0a soft.com

00000000 48 54 54 50 2f 31 2e 31 20 32 30 36 20 50 61 72 HTTP/1.1 206 Par
00000010 74 69 61 6c 20 43 6f 6e 74 65 6e 74 0d 0a 43 6f tial Content..Co
00000020 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 nnection : keep-a
00000030 6c 69 76 65 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65 live..Content-Le
00000040 6e 67 74 68 3a 20 31 30 34 38 35 37 36 0d 0a 43 ngth: 10 48576..C
00000050 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 61 70 70 Content-Type: app
00000060 6c 69 63 61 74 69 6f 6e 2f 6f 63 74 65 74 2d 73 lication/octet-s
00000070 74 72 65 61 6d 0d 0a 43 61 63 68 65 2d 43 6f 6e tream..Cache-Con
00000080 74 72 6f 6c 3a 20 70 75 62 6c 69 63 2c 20 6d 61 trol: public, ma
00000090 70 2d 61 67 65 3d 31 37 32 30 30 30 30 0d 0a x-age=17 200000..
000000A0 4d 53 2d 43 6f 72 72 65 6c 61 74 69 6f 6e 49 64 MS-CorrelationId
000000B0 3a 20 65 36 31 30 61 37 64 62 2d 38 66 63 35 2d : e610a7db-8fc5-
000000C0 34 36 33 33 2d 39 30 64 65 2d 30 65 30 38 37 39 4633-90de-0e0879
000000D0 61 32 61 34 37 63 0d 0a 4d 53 2d 52 65 71 75 65 a2a47c...MS-Reque
000000E0 73 74 49 64 3a 20 37 39 31 37 31 33 36 32 2d 63 stId: 79 171362-c
000000F0 31 34 31 2d 34 35 66 33 2d 39 33 31 34 2d 32 63 141-45f3-9314-2c
00000100 38 62 35 34 66 36 62 39 61 35 0d 0a 4d 53 2d 43 8b54f6b9a5..MS-C
00000110 56 3a 20 6a 55 4f 59 75 59 49 78 6d 55 79 59 70 V: yUOyU YXmUyYp
00000120 48 6f 59 2e 30 2e 32 2e 31 2e 30 2e 30 2e 39 2e HoY.0.2. 1.0.0.9.
00000130 31 2e 31 2e 32 2e 32 2e 31 2e 31 2e 30 0d 0a 43 1.1.2.2. 1.1.0..C
00000140 6f 6e 74 65 6e 74 2d 44 69 73 70 6f 73 69 74 69 Content-Disposition:
00000150 6f 6e 3a 20 61 74 74 61 63 6d 6d 65 6e 74 3b 20 on: attachment;
00000160 66 69 6c 65 6e 61 6d 65 3d 4d 69 63 72 6f 73 6f filename="Microso
00000170 66 74 2e 59 6f 75 72 50 68 6f 6e 65 5f 31 2e 32 ft.YourP hone_1.2
00000180 36 30 37 32 2e 32 35 35 2e 30 5f 6e 65 75 74 72 6072.255..0 neutr
00000190 61 6c 5f 7e 5f 38 77 65 6b 79 62 33 64 38 62 62 al_..8we kyb3d8bb
000001A0 77 65 2e 41 70 70 78 42 75 6e 64 6c 65 0d 0a 4c we-AppxB undle..L
000001B0 61 73 74 2d 4d 6f 64 69 66 69 65 64 3a 20 54 68 ast-Modified: Th
000001C0 75 2c 20 31 30 20 53 65 70 20 32 30 32 36 20 32 u, 10 Sep 2026 2
000001D0 32 3a 30 38 3a 33 30 20 47 4d 54 0d 0a 45 54 61 2:08:30 GMT..ETa
000001E0 67 3a 20 22 62 76 4e 37 4f 31 4d 42 6c 4e 77 44 g: "bvH7 01MB1mW0
000001F0 46 4a 48 79 53 54 59 76 6a 51 57 4c 42 5a 59 3d fJhy5TYv jQhL8Zy
00000200 22 0d 0a 41 63 63 65 70 74 3d 52 61 6e 67 65 73 ".Accept-Ranges
00000210 3a 20 62 79 74 65 73 6d 0a 4d 61 74 65 3a 20 54 .Content-Type: T

34 client ptes, 3 456 server ptes, 67 turns.

Entire conversation (22 MB) Show as Hex Dump No delta times Stream 94

Find: Case sensitive Find Next

Filter Out This Stream Print Save as... Back Close Help

Wireshark - Export - HTTP object list

Text Filter: Content Type: All Content-Types

Packet	Hostname	Content Type	Size	Filename
589	msedge.b.tlu.dl.delivery.mp.microsoft.com	application/x-chrome-extension	487 kB	68591036-2289-4858-9f7f-9149e89c8a087P1=1789758789&P2=404&P3=2&P4=jTD971eYxVmq5JfUJlW2epNkAUoCA6og5z%2fsGAfZ4xz1jZMjUj5YBk%
1066	msedge.b.tlu.dl.delivery.mp.microsoft.com	application/x-chrome-extension	273 kB	68591036-2289-4858-9f7f-9149e89c8a087P1=1789758789&P2=404&P3=2&P4=jTD971eYxVmq5JfUJlW2epNkAUoCA6og5z%2fsGAfZ4xz1jZMjUj5YBk%
1409	msedge.b.tlu.dl.delivery.mp.microsoft.com	application/x-chrome-extension	331 kB	68591036-2289-4858-9f7f-9149e89c8a087P1=1789758789&P2=404&P3=2&P4=jTD971eYxVmq5JfUJlW2epNkAUoCA6og5z%2fsGAfZ4xz1jZMjUj5YBk%
1786	dl.delivery.mp.microsoft.com	application/octet-stream	14 kB	pieceshash
1797	dl.delivery.mp.microsoft.com	application/octet-stream	2362 bytes	pieceshash
1818	dl.delivery.mp.microsoft.com	application/octet-stream	1187 bytes	pieceshash
2375	tlu.dl.delivery.mp.microsoft.com	application/octet-stream	336 kB	dff8b86fe-1d0b-437d-ac77-708cee3a386e7P1=1789651680&P2=404&P3=2&P4=AD5i3AE2h6oiSkUw4%2f8YhqmynZap1a47MqyJ1hUPcxW%2f2fnoar
2776	tlu.dl.delivery.mp.microsoft.com	application/octet-stream	441 kB	de2aac7e-1ffa-4a30-bf26-8afcacbb98707P1=1789646675&P2=404&P3=2&P4=HU9FXtOuKMuVv3xh05gbrJ9JZpL%2fxtfMTpBFRxCjr8Mz2F%2fJNNAI
2869	download.windowsupdate.com	application/vnd.ms-cab-compressed	7321 bytes	46007954_c343635ce2a5f5ceac82727c24060f5d3f4463.cab
3022	download.windowsupdate.com	application/vnd.ms-cab-compressed	7315 bytes	46007954_7b27af726f1b0a7352e4ee7e37075d9c283a773.cab
3239	download.windowsupdate.com	application/vnd.ms-cab-compressed	7263 bytes	46007511_118065666ed2ae1536d3e8a8a6c51d7b8819c31.cab
3520	download.windowsupdate.com	application/vnd.ms-cab-compressed	7257 bytes	46007510_bd03da1f4aa58366c784cf4e935f7acffa4e0a44.cab
3527	tlu.dl.delivery.mp.microsoft.com	application/octet-stream	854 kB	a127317b-857e-4d28-9483-ab35886413117P1=1789646218&P2=404&P3=2&P4=KUKPWsOQeGloO3AmFBY5sd0VAs53hT3UPPpCnV%2f2bnCtzBq4XQ%
3549	download.windowsupdate.com	application/vnd.ms-cab-compressed	7259 bytes	46007509_9fdbf6d98eae1e392fa1d38105dd472a30370ff.cab
3576	download.windowsupdate.com	application/vnd.ms-cab-compressed	7265 bytes	46007508_0df4a2c0b413483839bd1a3994d251a6c6e3d30e.cab
3599	download.windowsupdate.com	application/vnd.ms-cab-compressed	7263 bytes	46007507_f323d9968deb7f14470cb18054fbc81cb0245c9.cab
3620	download.windowsupdate.com	application/vnd.ms-cab-compressed	7259 bytes	46007506_8d4c844cf328c09a9a59866041bd312a81bd.cab
3653	download.windowsupdate.com	application/vnd.ms-cab-compressed	7259 bytes	46007505_ac926c82cb1239f91198616249f4aab8740210d.cab
3683	download.windowsupdate.com	application/vnd.ms-cab-compressed	7265 bytes	46007504_4913db94e0eff014b2b2e53c425001cacb7b6b17.cab
3723	download.windowsupdate.com	application/vnd.ms-cab-compressed	7263 bytes	46007503_f0d6050402256a94c9ae378b9e0cd352bb307c6a.cab
3754	download.windowsupdate.com	application/vnd.ms-cab-compressed	7265 bytes	46007502_bb15f732a7553f10596524f55b1115e12eebbac5.cab
3773	download.windowsupdate.com	application/vnd.ms-cab-compressed	7259 bytes	46007501_ddc6305d672a67f5c88c7209048b613c47d8da1f.cab
3774	download.windowsupdate.com	application/vnd.ms-cab-compressed	7261 bytes	46007500_68aad0ddfd2a6d245a54a2b107e011cc8069438f.cab
3785	download.windowsupdate.com	application/vnd.ms-cab-compressed	7265 bytes	46007499_c455c468f20bd96d1a330813b452deda1e040b78.cab
3796	download.windowsupdate.com	application/vnd.ms-cab-compressed	7263 bytes	46007498_a6f98bb6ab934ad83825cee7fdd9269b404f2047.cab
3820	download.windowsupdate.com	application/vnd.ms-cab-compressed	7267 bytes	46007497_88e080bf1a1fa70118fc22fc3404f3ab48f69d2.cab
3835	download.windowsupdate.com	application/vnd.ms-cab-compressed	7257 bytes	46007451_f008e537bc5e4c6e20006509f1cede38245cbb.cab
3848	download.windowsupdate.com	application/vnd.ms-cab-compressed	7261 bytes	46007450_9d23d0fd1e94dfdbd39de46ecfa79ae1aee59de6.cab
3875	download.windowsupdate.com	application/vnd.ms-cab-compressed	7261 bytes	46007449_f2665cfa18c9895c4c84d7af624c756aa5888d9.cab
3892	download.windowsupdate.com	application/vnd.ms-cab-compressed	7263 bytes	46007448_cf0c9db38f2b2c3b14f3147276c3faa31e444.cab
3977	download.windowsupdate.com	application/vnd.ms-cab-compressed	7263 bytes	46007447_d439524adea524276678d35a6e2a82e123923be.cab
4043	download.windowsupdate.com	application/vnd.ms-cab-compressed	7267 bytes	46007446_f25bfd79d6b3a7111768314dbdff9a6b14102e.cab

Save Save All Preview Close Help

The image shows a Wireshark packet capture analysis. The top pane displays a list of packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The middle pane shows the details of the selected packet (No. 18038), including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
11998	0.000004180	192.168.0.53	140.248.134.172	TCP	66	[TCP Dup ACK 11980#1] 60213 → 80 [ACK] Seq=889 Ack=1149509 Win=524032 Len=0 SLE=1153829 SRE=1156709
11997	0.000006800	192.168.0.53	140.248.134.172	TCP	66	[TCP Dup ACK 11992#1] 60211 → 80 [ACK] Seq=1333 Ack=2485430 Win=1048320 Len=0 SLE=2502710 SRE=2505590
11995	0.000003600	140.248.134.172	192.168.0.53	TCP	1494	[TCP Previous segment not captured] 80 → 60213 [ACK] Seq=1153829 Ack=889 Win=148480 Len=1440 [TCP PDU reassembled in 11213]
11993	0.001422000	140.248.134.172	192.168.0.53	TCP	1494	[TCP Previous segment not captured] 80 → 60211 [ACK] Seq=2502710 Ack=1333 Win=149504 Len=1440 [TCP PDU reassembled in 11213]
11991	0.000001800	192.168.0.53	140.248.134.172	TCP	66	[TCP Dup ACK 11979#1] 60214 → 80 [ACK] Seq=889 Ack=1290627 Win=524032 Len=0 SLE=1294947 SRE=1296387
11990	0.078805900	140.248.134.172	192.168.0.53	TCP	1494	[TCP Previous segment not captured] 80 → 60214 [PSH, ACK] Seq=1294947 Ack=889 Win=148480 Len=1440 [TCP PDU reassembled in 11213]
11697	0.668506800	192.168.0.53	52.123.128.14	TCP	55	[TCP Keep-Alive] 49394 → 443 [ACK] Seq=3494 Ack=573 Win=253 Len=1
10838	0.000003900	140.248.134.172	192.168.0.53	TCP	1494	[TCP Out-Of-Order] 80 → 60211 [ACK] Seq=1821988 Ack=889 Win=148480 Len=1440 [TCP PDU reassembled in 11213]
10835	0.000002000	140.248.134.172	192.168.0.53	TCP	1494	[TCP Out-Of-Order] 80 → 60211 [PSH, ACK] Seq=1820548 Ack=889 Win=148480 Len=1440 [TCP PDU reassembled in 11213]
10834	0.007103300	140.248.134.172	192.168.0.53	TCP	1494	[TCP Out-Of-Order] 80 → 60211 [ACK] Seq=1819108 Ack=889 Win=148480 Len=1440 [TCP PDU reassembled in 11213]
10832	0.000006200	140.248.134.172	192.168.0.53	TCP	1494	[TCP Out-Of-Order] 80 → 60211 [PSH, ACK] Seq=1817668 Ack=889 Win=148480 Len=1440 [TCP PDU reassembled in 11213]
10831	0.004239500	140.248.134.172	192.168.0.53	TCP	1494	[TCP Out-Of-Order] 80 → 60211 [ACK] Seq=1816228 Ack=889 Win=148480 Len=1440 [TCP PDU reassembled in 11213]
10829	0.000001200	140.248.134.172	192.168.0.53	TCP	1494	[TCP Out-Of-Order] 80 → 60211 [PSH, ACK] Seq=1814788 Ack=889 Win=148480 Len=1440 [TCP PDU reassembled in 11213]
10828	0.000001300	140.248.134.172	192.168.0.53	TCP	1494	[TCP Out-Of-Order] 80 → 60211 [PSH, ACK] Seq=1813348 Ack=889 Win=148480 Len=1440 [TCP PDU reassembled in 11213]
10827	0.000001400	140.248.134.172	192.168.0.53	TCP	1494	[TCP Out-Of-Order] 80 → 60211 [PSH, ACK] Seq=1811908 Ack=889 Win=148480 Len=1440 [TCP PDU reassembled in 11213]

Frame 18038: Packet, 1494 bytes on wire (11952 bits), 1494 bytes captured (11952 bits) on interface \Device\NPF_{C0A5F3B3-222D-4FDF-8190-A5F355C286F5}, id 0
 Ethernet II, Src: Routerboardc_76:02:63 (e4:8d:8c:76:02:63), Dst: Intel_d4:e8:01 (00:d4:9e:d4:e8:01)
 Internet Protocol Version 4, Src: 140.248.134.172, Dst: 192.168.0.53
 Transmission Control Protocol, Src Port: 80, Dst Port: 60211, Seq: 1821988, Ack: 889, Len: 1440

TCP Analysis Flags: Label

Packets: 95118 · Displayed: 3581 (3.8%) · Dropped: 0 (0.0%)

The Conclusion of my investigation is that port 80 was confirmed to be a legitimate Windows update, Microsoft Delivery Optimization and Microsoft Edge extension-update activity – not encrypted tunnelling or anomalous behaviour. The TCP error rate (3.8%) is consistent with a Normal Wi-Fi contention during a heavy bulk download period rather than an underlying network fault, as the flagged packets are dominated by benign categories (out-of-order, duplicate ACK) rather than signs of actual data loss (zero-window, repeated fast retransmissions).